

Chapter 1

A. Questions

1. Please explain the three major design purposes of the operating system ◦

Answer: Simplicity, efficiency, and evolution

2. What are the 4 main functions of the operating system?

Answer:

- (a) Trip management
- (b) Memory management
- (c) File management
- (d) I/O system management

3. What is the main job of a CPU? How many tasks can a CPU perform at once?

Answer:

- (a) Specializes in various calculations and is responsible for communicating with peripherals.
- (b) One tasks ◦

4. What are the steps involved in the execution of the command?

Answer:

Command extraction, command decoding, data extraction, command execution, and data storage.

B Advanced questions

1. Please follow the instructions below to draw the final contents of the register when the command of memory addresses 200 to 206 is executed ◦

Instruction format :

opcode(4-bit)	memory address(12-bit)
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List of opcodes :

opcode	Description	opcode	Description
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0000	Load data from memory into register A	0010	Subtract the values of registers A and B and put them into register A
0001	Loads data from memory into register B	0011	Writes the value in register A to memory

Memory initial content :

Registers (hexadecimal)		Memory (hexadecimal)	
	PC	200	0 4 0 2
	IR	202	1 4 0 0
	Register A	204	2 0 0 0
	Register B	206	3 4 0 0
			
		400	0 0 0 2
		402	0 0 1 6

Answer :

register (HEX)		Memory (hex)	
	PC	200	0 4 0 2
	IR	202	1 4 0 0
	Register A	204	2 0 0 0
	Register B	206	3 4 0 0
			
		400	0 0 1 4
		402	0 0 1 6